

Path of Exile Economy Analysis

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Abstract—This project utilizes the Path of Exile economy from the Mercenaries League to analyze the changes of a simulated economy. Using poe.ninja, 43,126 rows of economy data over a 4 month period are acquired and analyzed to identify trends and large shifts.

I. DATASET DESCRIPTION

- Mercenaries League Economy. The data is inspired by the Stack Overflow data dumps, licensed as cc-by-sa.
- <https://poe.ninja/poe1/data>. Free open download.
- The currency portion of the data is 2MB being, 6 columns and 43,126 rows tracking daily values of currency items relative to other currency items. I started by focusing the data to only show when spending Chaos orbs. This lowered the data to 26,906 rows.
- Contains daily snapshots of each individual currency item and its relative value in comparison to 'Chaos Orbs' (the general community agreed upon trading currency). Furthermore, it contains a confidence column to help eliminate unclear data and evidence of price-fixing in the market.

A. Dataset Example

TABLE I

SAMPLE CURRENCY ECONOMY DATA FROM POE.NINJA (ABYSS LEAGUE)

League	Date	Get	Pay	Value
Abyss;	2017-12-09;	Exalted Orb;	Chaos Orb;	26.16644;
Abyss;	2017-12-10;	Exalted Orb;	Chaos Orb;	35.92556;
Abyss;	2017-12-11;	Exalted Orb;	Chaos Orb;	43.5739;
Abyss;	2017-12-12;	Exalted Orb;	Chaos Orb;	47.99698;
Abyss;	2017-12-13;	Exalted Orb;	Chaos Orb;	45.45841;
Abyss;	2017-12-14;	Exalted Orb;	Chaos Orb;	41.32816;

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II. DISCOVERY QUESTIONS

A. How do different currencies trend up or down together?

This question examines whether groups of currencies exhibit similar price movements over the course of the league. By analyzing correlations and clustering currency time-series data, this analysis aims to identify currencies that rise and fall together, potentially demonstrating shared gameplay functions, crafting dependencies, or common demand drivers. Understanding these co-movement patterns provides insight into the structural relationships within the in-game economy.

B. How do currency value trends evolve over the course of a league, and what distinct behavioral phases can be identified?

This question examines how currency value trends change throughout the league and whether distinct economic phases appear over time. By analyzing price movements and grouping currencies with similar patterns, the analysis aims to identify shifts in the economy as player progression, crafting, and endgame activities evolve. Recognizing these patterns helps reveal how the in-game economy changes during the league lifecycle.

C. Which currencies behave as outliers compared to overall market trends?

This question aims to identify currencies whose value behavior deviates significantly from the general market trends observed during the league. By comparing individual currency trajectories against aggregate market movement, the analysis highlights outliers characterized by unusually high volatility, abrupt price changes, or sustained divergence. These currencies may reflect speculative activity, sudden shifts in player demand, or the impact of game updates and league mechanics.

III. PLANNED DATA MINING TECHNIQUES

The goal of the planned techniques is to explore patterns and relationships in the in-game currency economy, rather than to predict future prices. Since this is an exploratory study, the focus is on methods that can reveal trends, groups, and unusual behavior among currencies over time.

Before applying any analysis, the data will be cleaned, normalized, and organized so that each currency is represented as a time-series of daily values. This allows us to compare currencies and track changes throughout the league.

Clustering and dimensionality reduction are the main techniques. Clustering will group currencies that behave similarly, showing which rise or fall together. Dimensionality reduction (PCA) will simplify the data and make it easier to visualize overall trends and patterns. Together, these methods provide a clear view of how the video game economy evolves.

A. Clustering

Clustering will be used to group currencies that show similar patterns over time. By looking at how each currency's value changes daily, we can identify sets of currencies that rise and fall together or behave in similar ways. This helps highlight

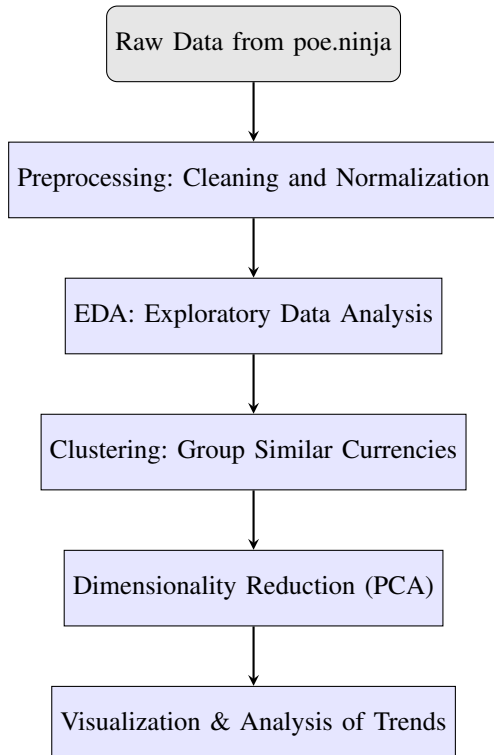
early-game versus late-game currencies, stable baseline currencies, and more volatile or unusual ones. Clustering provides a way to see structure in the market without assuming anything about the outcome. Two clustering techniques were applied to identify patterns in the Path of Exile currency market data. K-Means clustering was used on individual trade records using the features exchange value and confidence level. The data was standardized before clustering, and $k = 3$ clusters were used to group trades into similar economic states. This clustering was visualized against the week of the league to observe how trade values evolve over time. Furthermore, agglomerative hierarchical clustering was used to group currencies based on how their values changed over the course of the league. Instead of clustering individual trades, this method analyzed overall currency behavior. For each currency, features were created including early-league value, late-league value, percent change, and price volatility. Together, these clustering approaches reveal currencies that share similar price trends, volatility, and economic trajectories within the in-game market.

B. Dimensionality Reduction (PCA)

Principal Component Analysis (PCA) will be used to simplify the dataset and make patterns easier to see. Each currency has many daily values, so PCA reduces this high-dimensional data to just a few key components that explain most of the variation. This helps visualize overall trends, understand what drives changes in currency values, and support the clustering results by showing how currencies are related in a lower-dimensional space.

IV. PLANNED ANALYSIS PIPELINE

A. Flowchart



B. M2: Initial Implementation - March 5th

- Perform exploratory data analysis (EDA) on the currency dataset, including visualizations of trends and distributions.
- Conduct a first attempt at pattern discovery using either clustering or association rule mining.
- Document preliminary findings and observations for further analysis in later milestones.

C. M3: Complete Implementation - April 2nd

- Apply multiple data mining techniques (clustering, PCA, and optionally anomaly detection) to the dataset.
- Compare and analyze patterns discovered by different techniques.
- Ensure code quality, proper documentation, and reproducibility of results.
- Refine interpretation of findings based on observed trends and relationships.

D. M4: Final Deliverable - May 3rd

- Produce a polished and fully reproducible analysis.
- Deliver meaningful interpretations of discovered patterns in the currency economy.
- Submit a portfolio-ready repository containing clean code, analysis notebooks, and visualizations.
- Prepare a final presentation or demonstration video summarizing results.

E. Anticipated Challenges

- Ensuring high-quality preprocessing and handling missing or inconsistent data.
- Interpreting patterns in noisy or volatile currency value trends.
- Balancing multiple techniques and integrating their results into a coherent analysis.
- Maintaining reproducibility and clarity in both code and visualizations for final submission.

V. EDA - EXPLORATORY DATA ANALYSIS

A. Dataset Overview

- There are 6 columns of data.
- There are 43,126 rows of data that I culled to 26,906 rows.
- These rows encompass the exchange rates of 198 different unique currencies and their value ratio to chaos per day over the course of the 4 month league.
- Using the Confidence column we can see that 94.4% of the data points are classified as 'High' confidence so generally the data is trustworthy and therefore I decided to keep all the data.
- There are no missing data and no instances of fully identical data so that was not something I had to deal with.

B. Preprocessing and Simplification

- I did a conversion of Date to datetime and extraction of temporal features: day, month, week. Additionally, created bounds for early league and late league based upon lower than and higher than the median week.
- The dataset was filtered to use 'Chaos Orb' as the standard pay currency to simplify value and economy translations.
- Mapped confidence to numeric values for analysis (Low=1, Medium=2, High=3).
- Used a whitelist for currency correlations.

C. Average Chaos Exchange Rate Over Time

Shows that the relative value of chaos to all other currencies does change and trends upwards relatively over time.

D. Correlation Matrix (Selected Key Currencies)

In asking my first discovery question, I created this matrix to understand the currency relations for a few popular whitelisted currencies. We can see that high value currencies generally have a high positive correlation. Here are some notable correlations and possible explanations:

- Divine Orb to Grand Eldritch Ichor: 0.95 – These currencies have a very high positive correlation because they are both heavily used in late-game crafting. Players tend to accumulate them as valuable resources for high-level item upgrades, which causes their prices to rise together over the league. The demand patterns for these currencies are closely linked, reflecting the timing and intensity of late-game economic activity.
- Divine Orb to Ancient Orb: 0.87 – These two currencies also trend up together due to their strong association with late-game crafting and high-risk gambling mechanics. Both are sought after for enhancing powerful items or attempting high-stakes upgrades, resulting in synchronized price increases. While the correlation is slightly lower than with Grand Eldritch Ichor, the shared economic role in advanced gameplay explains the similar market behavior.
- Orb of Fusing to Reflecting Mist: -0.61 - They have a negative correlation likely due to one having high value and need early while the other can't serve much current purpose while the first is essentially unneeded late and the second is crucial to high investment late game crafts.
- Orb of Alchemy to Veiled Chaos Orb: -0.64 - They have a negative correlation likely because one currency is highly valuable and primarily needed early in the league, while the other becomes essential for high-end crafting later on, making the first less useful in the late game.

E. Selected Currencies: Early vs Late League

We can see that on average the most expensive currencies will be more expensive later in the league than early. Furthermore, the cheapest currencies see a sharp drop in value later in the league.

F. Currencies Deviating Most from Market Average

Here we can see that the 'Mirror of Kalandra' has by far the greatest value of all currency items in the game and is therefore very rare and powerful.

VI. CLUSTER RESEARCH

A. Value vs Week Cluster

Through this graph we can see that as the game progresses more items become increasingly valuable in comparison to Chaos and it creates a cluster starting a few weeks into the league of significantly more valuable currencies than early.

B. Agglomerative Clustering: Early vs Late Values

Agglomerative hierarchical clustering was applied to group currencies based on their overall behavior during the league. For each currency, features were constructed from early-league value, late-league value, percent change, and price volatility. Features were log-transformed and standardized to reduce skew and ensure balanced clustering. Using Ward linkage with Euclidean distance, currencies with similar trends and volatility were grouped into three clusters. The results were visualized in a scatter plot comparing early and late values, highlighting groups of currencies that behave similarly in the in-game market.

VII. KEY DATA MINING INSIGHTS

The exploratory analysis revealed clear patterns and relationships in the Path of Exile currency economy. Correlation analysis showed that high-value currencies used in late-game crafting, such as Divine Orbs and Grand Eldritch Ichor, tend to rise together, while certain early-game currencies exhibit negative correlations with late-game focused currencies. Clustering methods provided additional insight into market behavior: K-Means clustering on individual trade records identified groups of trades with similar value and confidence levels, showing how currency valuations evolve week by week. Agglomerative hierarchical clustering of currencies based on early- and late-league value, percent change, and volatility further highlighted distinct groups of currencies that share similar economic trajectories throughout the league. Outlier analysis revealed rare or highly volatile currencies, such as Mirror of Kalandra, which deviate significantly from market trends. Together, these analyses provide a comprehensive view of the in-game economy, uncovering currency co-movement, phase-dependent behaviors, and unusual patterns that reflect both player demand and game mechanics.

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